

Current SoCs

TI OMAP3430

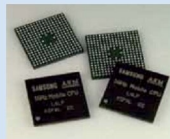
TI OMAP3430 features an ARM Cortex A8 processor clocked at 600 MHz. ARM Cortex A8 processor is based on the ARMv7 architecture and is claimed to scale in speed from 600 MHz to greater than 1 GHz. In fact, many people run hacked kernels where the processor is overclocked from 600 MHz to 1.15 MHz. The downside, however, is that the SoC uses a lot of power as its clock gets scaled up.

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Samsung Hummingbird 1GHz

Samsung Hummingbird too is based on the ARM Cortex A8 architecture, albeit at a smaller 45nm process than the



TI OMAP 3430. It was jointly developed along with Intrinsity. Intrinsity tweaked the CPU by trimming inefficiencies in the way logic gates are used by using 1-of-n domino logic (NDL). Due to this, the Hummingbird CPU delivers not only

high media and data crunching performance in mobile devices but lowered power consumption as well.

Hummingbird comes with 32 kB of both data and instruction cache, a variable-size L2 memory cache and the ARM NEON multimedia extension. This SoC features a PowerVR SGX 540 GPU.

Apple A4 800MHz

Apple A4 SoC is based on the ARM architecture. It is designed by Apple and manufactured by Samsung. It combines an ARM Cortex-A8 CPU with a PowerVR GPU and emphasises power efficiency. The chip made a commercial debut with the release of Apple's iPad tablet, followed shortly by the



iPhone 4 smartphone, the fourth-generation iPod Touch and the second-generation Apple TV.

The A4 processor package does not contain RAM but supports point-of-purchase (PoP) installation. This allows

the different devices which utilise it to have different RAM configurations, like 256MB low-power DDR SDRAM for the iPad and 512MB low-power DDR SDRAM for the iPhone 4. The SoC features a PowerVR SGX 535 GPU.

NVIDIA Tegra 2 1GHz

The NVIDIA Tegra 2 is a dual-core SoC that features an ultra-low-power (ULP) NVIDIA GeForce GPU with four pixel shaders and four vertex shaders. The processor within is actually a dual-core ARM Cortex-A9 CPU. The Cortex-A9 is 25 per cent Dhrystone MIPS faster than Cortex-A8. Cortex-A9 runs 1 GHz per core and has a 1MB L2 cache as well as one 32kB L1 cache per core. The GPU is a fully programmable OpenGL ES 2 supporting processor with eight cores. Tegra 2 being a NVIDIA product, a lot of Android games are believed to be optimised for it.



This chipset also features a 1080p playback processor, which allows HD movie playback without taxing the main processor and thus saving battery life. It has been designed using a 40nm process. There is also a version of this SoC supporting 3D displays. Named the Tegra 2 3D, this SoC uses a higher-clocked CPU (1.2GHz) and GPU.

Samsung Exynos 4210 1.2GHz

The Exynos 4210 uses the CortexA9 dual-core. It provides features such as dual-core CPU, high memory bandwidth,

native triple display (2 WSVGA+1 HDMI out simultaneously), 1080p video decode and encode hardware, 3D graphics hardware and high-speed interfaces such as SATA and USB. The application processor also supports DDR-based eMMC 4.4 interface to increase the file system's performance. Exynos 4210 uses ARM's Mali-400 MP GPU. This graphics GPU is a move away from the PowerVR GPU of the Samsung Galaxy S.



Qualcomm Snapdragon MSM8260 1.5GHz

Also known as the Snapdragon S3, this SoC features dual-core Scorpion CPUs and an Adreno 220 GPU built using a 45nm process. It uses an ARMv7 instruction set and has single-channel 333MHz ISM/266 MHz LPDDR2 and 512kB L2 cache. The Adreno GPU has often been